MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – ASSESSMENT TEST** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter: Areas Related to Circles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 11.
2. **Section A** comprises 5 Multiple Choice Questions (MCQs) of 1 mark each.
3. **Section B** comprises 3 Short Answer Type-I questions of 2 marks each.
4. **Section C** comprises 3 Short Answer Type-II questions of 3 marks each.
5. Use of calculators is not permitted. Take $\pi = \frac{22}{7}$ unless stated otherwise.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. **Area of the largest triangle that can be inscribed in a semi-circle of radius r units is:** [1]

(A) r^2 sq. units
(B) $\frac{1}{2}r^2$ sq. units
(C) $2r^2$ sq. units
(D) $\sqrt{2}r^2$ sq. units

Q2. **If the perimeter of a circle is equal to that of a square, then the ratio of their areas is:** [1]

(A) 22 : 7
(B) 14 : 11
(C) 7 : 22
(D) 11 : 14

Q3. **It is proposed to build a single circular park equal in area to the sum of areas of two circular parks of diameters 16 m and 12 m in a locality. The radius of the new park would be:** [CBSE 2012] [1]

(A) 10 m
(B) 15 m
(C) 20 m
(D) 24 m

Q4. **If the areas of two circles are in the ratio 4 : 9, then the ratio of the perimeters of their semi-circles is:** [1]

(A) 2 : 3
(B) 3 : 2
(C) 1 : 2
(D) 1 : 3

Q5. **If the difference between the circumference and the radius of a circle is 37 cm. If $\pi = \frac{22}{7}$, then the circumference (in cm) of the circle is:** [1]

(A) 15.4
(B) 44
(C) 14
(D) 7

SECTION B: Short Answer Type-I (3 × 2 = 6 Marks)

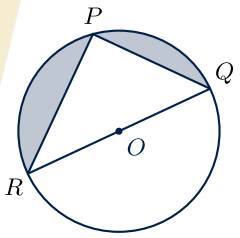
Q6. A bicycle wheel makes 5000 revolutions in moving 11 km. Find the diameter of the wheel. [2]

Q7. The perimeter of a sector of a circle of radius 5.6 cm is 27.2 cm. Find the area of the sector. [2]

Q8. A chord AB of a circle of radius 10 cm subtends a right angle at the centre. Find the area of the major and minor segments. [Take $\pi = 3.14$] [NCERT] [2]

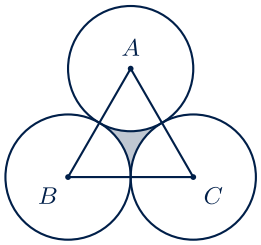
SECTION C: Short Answer Type-II (3 × 3 = 9 Marks)

Q9. Two circles touch externally. The sum of their areas is 130π sq cm and the distance between their centres is 14 cm. Find the radii of the circles. [3]



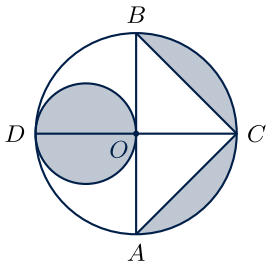
Q10. Find the area of the shaded region in the given figure, if $PR = 24$ cm, $PQ = 7$ cm and O is the centre of the circle. [3]

[CBSE 2008]



Q11. The area of an equilateral triangle is 17320 cm^2. With each vertex of the triangle a circle is drawn with radius equal to half the length of the side of the triangle in the given figure. Find the area of the shaded region. [Take $\sqrt{3} = 1.732$] [3]

— OR —



In the given figure, AB and CD are two diameters of a circle (with centre O) perpendicular to each other and OD is the diameter of the smaller circle. If $OA = 7$ cm, find the area of the shaded region. [3]

[NCERT]

★★★ END OF QUESTION PAPER · ALL THE BEST ★★★